Problem Statement

Design a feasible handheld mechanical nutcracker for Macadamia nuts.

Objectives

- Determine neccesary dimensions for level based nutcracker
- Have sufficent Mechanical Advantage to crack macadamia nut
- Design a usable Nutcracker machine with Linear actuator

Parameters

- $f_{crack} = 2179.858N$
- f_{grip}: 330N
- Nut Diameter: 25.4mm
- Actuator force fact: 667

Approach

- Geometry simplified
- friction negligible
- level base nutcracker

Design Calculation

- $MA = \frac{f_{crack}}{f_{grip}} = \frac{2179.858}{330N} = 6.604$

- Length

$L_2 = L_1 \cdot MA = 45mm \cdot 6.604 = 297mm = 11.6in$

Acuator Modifications

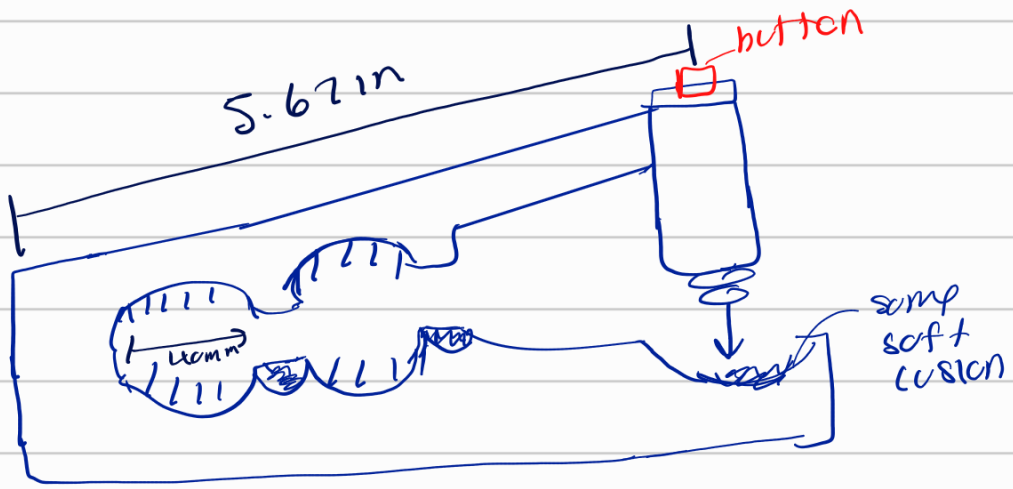
- New MA

$$MA_{new} = \frac{2179.858}{667N} = 3.2$$

- New Length:

$L_2 = L_1 \cdot MA = 45mm \cdot 3.2 = 144mm = 5.67in$

Usability

- Manual Design:
  - ~11.6in size, can fit within Drawer
  - 330N force feasible for human, maybe a bit large/difficult for human hand
  - force exerted may not be constant

- Actuator Design
  - Length is reduce to half
  - Distance travel for force is limited
  - Removes strength requirement